

Galois group of  $(x^2-2)(x^2-3)$  over  $\mathbb{Q}$ .

Since the polynomial  $(x^2-2)(x^2-3)$  has four distinct roots  $\pm\sqrt{2}, \pm\sqrt{3}$ , it is separable.

And, since  $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$ ,  $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{3})] \cdot [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 4$ .

So,  $|\text{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 4$  where  $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ , the splitting field over  $\mathbb{Q}$ .

Since the automorphism leaving  $\mathbb{Q}$  fixed must be conjugate the roots,

there are four possible automorphisms:

$$\begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}.$$

Let  $\sigma : K \rightarrow K; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$

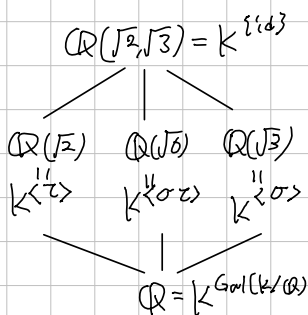
$\tau : K \rightarrow K; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}$ .

(That is,  $\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$  and  $\tau : \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$ , and we can observe that

$$\sigma\tau : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \quad \tau\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \quad \text{so } \sigma^2 = \tau^2 = 1 \text{ and } \sigma\tau = \tau\sigma$$

Now,  $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ , corresponding  $\sigma$  to  $(1, 2)$  and  $\tau$  to  $(2, 1)$ .

Moreover,



Note:  $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$  because: if  $\sqrt{2} = a + b\sqrt{3}$  for some  $a, b \in \mathbb{Q}$ , then  $2 = a^2 + 3b^2 + 2ab\sqrt{3}$ .  
it is Contradiction

Galois group of  $x^3 - 2$  over  $\mathbb{Q}$ .

Denote  $\zeta_3 = e^{i\frac{2\pi}{3}} = \frac{1}{2}(-1 + i\sqrt{3})$ , the 3<sup>th</sup> primitive root of unity.

Since  $\zeta_3^3 = 1$ , the set  $\{\sqrt[3]{2}, \sqrt[3]{2}\zeta_3, \sqrt[3]{2}\zeta_3^2\}$  are all roots of  $x^3 - 2$  over  $\mathbb{Q}$ .  
So, the splitting field is  $K = \mathbb{Q}(\sqrt[3]{2}, \zeta_3) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$ .

Observe that: the minimal polynomial of  $\sqrt[3]{2}$  over  $\mathbb{Q}$  is  $x^3 - 2$ , and  
"  $\zeta_3$  over  $\mathbb{Q}$  is  $x^2 + x + 1$ .

And, since  $x^2 + x + 1$  is irreducible over  $\mathbb{Q}(\sqrt[3]{2})$ , (it cannot have root in  $\mathbb{Q}(\sqrt[3]{2})$ )

$$[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}(\sqrt[3]{2})] \cdot [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6.$$

the tower lemma gave. And, since  $x^3 - 2$  is separable,  $|\text{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 6$ .

Now, construct explicitly the  $\text{Aut}(K/\mathbb{Q})$ .

Since any automorphism of  $K$  leaving  $\mathbb{Q}$  fixed must conjugate the roots,

it must map  $\sqrt[3]{2}$  to  $\sqrt[3]{2}$  or  $\sqrt[3]{2}\zeta_3$  or  $\sqrt[3]{2}\zeta_3^2$

and maps  $\zeta_3$  to  $\zeta_3$  or  $\zeta_3^2$ .

So, there are exactly 6 possibilities. Consider two automorphisms

$\sigma$  satisfying  $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\zeta_3$  and  $\sigma(\zeta_3) = \zeta_3$

and  $\tau$  satisfying  $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$  and  $\tau(\zeta_3) = \zeta_3^2 = -1 - \zeta_3$ .

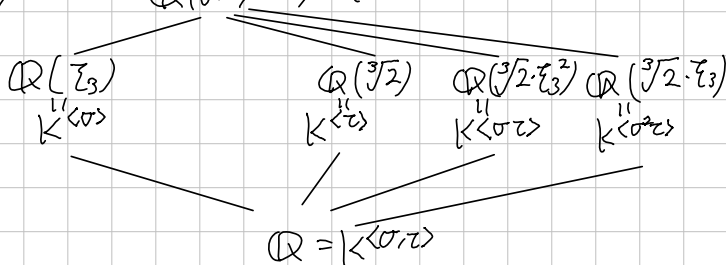
Then,  $\sigma^3: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3 \mapsto \sqrt[3]{2}\zeta_3^2 \mapsto \sqrt[3]{2}\zeta_3^3 = \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3 \end{cases}$ ,  $\tau^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \mapsto \zeta_3^4 = 1 - \zeta_3 = \zeta_3 \end{cases}$

are actually identities. Moreover,

$\sigma^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$ ,  $\sigma\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$ ,  $\sigma^2\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$ ,  $\tau\sigma: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$

So,  $\text{Aut}(K/\mathbb{Q}) = \{1, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} \cong S_3$ , corresponding  $\sigma$  to  $(1\ 2\ 3)$ ,  $\tau$  to  $(1\ 2)$

Moreover,  $\mathbb{Q}(\sqrt[3]{2}, \zeta_3) = K^{\langle \text{id} \rangle}$



How to find a fixed field corresponding to  $\langle \sigma\tau \rangle$ ?

As we know, the  $\text{Tr}_{K/K^{\langle \sigma\tau \rangle}}$  is invariant under any automorphism in  $\langle \sigma\tau \rangle$ .

So, Calculate:

$$\text{Tr}(\sqrt[3]{2}) = \text{id}(\sqrt[3]{2}) + (\sigma\tau)(\sqrt[3]{2}) = \sqrt[3]{2} + \sqrt[3]{2}\zeta_3 = \sqrt[3]{2}(1 + \zeta_3) = \sqrt[3]{2} \cdot (-\zeta_3^2).$$

Therefore,  $\sqrt[3]{2}\zeta_3^2$  is fixed by  $\text{id}$  and  $\sigma\tau$ , so  $\mathbb{Q}(\sqrt[3]{2}\zeta_3^2) \subseteq K^{\langle \sigma\tau \rangle}$ .

Moreover,  $[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}(\sqrt[3]{2}\zeta_3^2)] = 2$ , so the Galois theorem gives the result.

Galois group of  $x^8 - 2$  over  $\mathbb{Q}$ .

By the Eisenstein criterion,  $2 \in 2\mathbb{Z}$  but  $2 \notin 2^2\mathbb{Z}$ , so  $x^8 - 2$  is irreducible over  $\mathbb{Q}$ . And, the formal derivative  $8 \cdot x^7$  has only root as zero, but  $0^8 - 2 \neq 0$ , so the given polynomial is separable. The polynomial  $x^8 - 2$  has 8 distinct roots,

$$\sqrt[8]{2} \cdot \zeta_8^k \quad (k=0, 1, \dots, 7) \quad \text{where} \quad \zeta_8 = \exp(2\pi i \cdot \frac{1}{8}) = \frac{\sqrt{2}}{2}(1+i)$$

Therefore, the splitting field  $K = \mathbb{Q}(\sqrt[8]{2}, \dots, \sqrt[8]{2} \zeta_8^7) = \mathbb{Q}(\sqrt[8]{2}, i)$ , because the  $\subseteq$  inclusion is given by  $(\sqrt[8]{2})^4 = \sqrt{2}$ , so  $\zeta_8 \in \mathbb{Q}(\sqrt[8]{2}i)$ , and the  $\supseteq$  inclusion is given by  $\zeta_8^2 = \frac{1}{2}(2i) = i \in \mathbb{Q}(\sqrt[8]{2}, \dots, \sqrt[8]{2} \cdot \zeta_8^7)$ .

By the irreducibility, the minimal polynomial of  $\sqrt[8]{2}$  is  $x^8 - 2$ , and the minimal polynomial of  $i$  is  $x^2 + 1$  trivially. Since  $\mathbb{Q}(\sqrt[8]{2}) \subseteq \mathbb{R}$ ,  $i \notin \mathbb{Q}(\sqrt[8]{2})$ , thus the tower lemma gives

$$[K : \mathbb{Q}] = [\mathbb{Q}(\sqrt[8]{2}, i) : \mathbb{Q}(\sqrt[8]{2})] \cdot [\mathbb{Q}(\sqrt[8]{2}) : \mathbb{Q}] = 16.$$

Find explicitly  $\text{Gal}(K/\mathbb{Q})$ . Since an automorphism in  $\text{Gal}(K/\mathbb{Q})$  conjugates the  $\sqrt[8]{2}$  and  $i$ , therefore, we can define

$$\sigma: \begin{cases} \sqrt[8]{2} \mapsto \sqrt[8]{2} \cdot \zeta_8 \\ i \mapsto i \end{cases} \quad \text{and} \quad \tau: \begin{cases} \sqrt[8]{2} \mapsto \sqrt[8]{2} & (\sigma(\zeta_8) = \frac{1+i}{2} \cdot \sigma((\sqrt[8]{2})^4) = \zeta_8^5) \\ i \mapsto -i & (\tau(\zeta_8) = \tau(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i = \zeta_8^7) \end{cases}$$

Then, just simple calculation gives  $\text{id}, \sigma, \dots, \sigma^7, \tau, \tau\sigma, \dots, \tau\sigma^7$  are mutually distinct. So,  $\text{Gal}(K/\mathbb{Q}) = \{\text{id}, \sigma, \dots, \sigma^7, \tau, \dots, \tau\sigma^7\}$ . Further, observe that

$$\sigma^8 = \tau^2 = 1 \quad \text{and} \quad \sigma \cdot \tau: \begin{cases} \sqrt[8]{2} \mapsto \sqrt[8]{2} \cdot \zeta_8 \\ i \mapsto -i \end{cases}$$

$$\tau\sigma^3: \begin{cases} \sqrt[8]{2} \xrightarrow{0} \sqrt[8]{2} \cdot \zeta_8 \xrightarrow{1} \sqrt[8]{2} \cdot \zeta_8^6 \xrightarrow{1+5} \sqrt[8]{2} \cdot \zeta_8^7 \xrightarrow{1+30} \sqrt[8]{2} \cdot \zeta_8 \\ i \mapsto -i \end{cases}$$

Therefore  $\text{Gal}(K/\mathbb{Q}) = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma \cdot \tau = \tau\sigma^3 \rangle$ , the dicyclic group.

$$\text{Gal}(K/\mathbb{Q})$$

$$2 \mid$$

$$\langle \sigma \rangle$$

$$2 \mid$$

$$\langle \sigma^2 \rangle$$

$$2 \mid$$

$$\langle \sigma^4 \rangle$$

$$2 \mid$$

$$\{\text{id}\}$$

Galois group of  $x^5 - 3$  over  $\mathbb{Q}(\zeta)$ , where  $\zeta$  is a primitive fifth root of unity. Observe that:  $x^5 - 3$  has five distinct roots,  $\sqrt[5]{3} \cdot \zeta^k$  ( $k=0,1,\dots,4$ ).

Since the minimal polynomial of  $\zeta$  over  $\mathbb{Q}$  is  $x^4 + x^3 + x^2 + x + 1$ , therefore  $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$ . And, the minimal polynomial of  $\sqrt[5]{3}$  over  $\mathbb{Q}$  is  $x^5 - 3$ , therefore  $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] = 5$ . Therefore, the splitting field of  $x^5 - 3$  over  $\mathbb{Q}$ ,

$$K = \mathbb{Q}(\sqrt[5]{3}, \sqrt[5]{3}\zeta, \sqrt[5]{3}\zeta^2, \sqrt[5]{3}\zeta^3, \sqrt[5]{3}\zeta^4) = \mathbb{Q}(\sqrt[5]{3}, \zeta) = \mathbb{Q}(\sqrt[5]{3})\mathbb{Q}(\zeta)$$

has degree of 20 over  $\mathbb{Q}$ , that is,  $[K : \mathbb{Q}] = 20$ .

More precisely, by the tower lemma  $[K : \mathbb{Q}]$  is divided by  $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}]$  and  $[\mathbb{Q}(\zeta) : \mathbb{Q}]$ .

Since 5 and 4 are relatively prime,  $[K : \mathbb{Q}] \geq 20$ .

And,  $x^5 - 3 \in \mathbb{Q}(\zeta)[x]$  has  $\sqrt[5]{3}$  as a root, so

$$[\mathbb{Q}(\sqrt[5]{3}, \zeta) : \mathbb{Q}(\zeta)] \cdot [\mathbb{Q}(\zeta) : \mathbb{Q}] \leq 20.$$

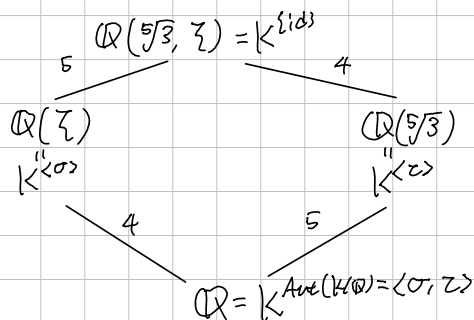
Consequently,  $[K : \mathbb{Q}] = 20$ . Now, by the Fundamental Theorem of Galois Theory,  $K/\mathbb{Q}(\zeta)$  is also Galois, with  $[K : \mathbb{Q}(\zeta)] = 5$ .

Therefore,  $\text{Gal}(K/\mathbb{Q}(\zeta))$  has order of 5, so it must be a cyclic group of order 5. Consequently, the automorphism

$$\sigma : \begin{cases} \sqrt[5]{3} \mapsto \sqrt[5]{3} \cdot \zeta \\ \zeta \mapsto \zeta \end{cases} \quad (\text{It is given by the conjugate isomorphism theorem}).$$

has order of 5, therefore  $\text{Gal}(K/\mathbb{Q}(\zeta)) = \langle \sigma \rangle \cong \mathbb{Z}_5$ .  $\square$

Latticisally,



where  $\tau : \begin{cases} \sqrt[5]{3} \mapsto \sqrt[5]{3} \\ \zeta \mapsto \zeta^2 \end{cases}$  Therefore,  $\text{Aut}(K/\mathbb{Q}) \cong \langle (1\ 2\ 3\ 4\ 5), (2\ 3\ 5\ 4) \rangle$ .

Galois group of  $x^6+3$  over  $F = \mathbb{Q}$  or  $\mathbb{F}_5$  or  $\mathbb{F}_7$ . (UCLA Quid, 2020 Fall)  
 For each case, determine the splitting field  $K$  and evaluate  $[K:F]$ .  
 And, find the Galois group  $\text{Gal}(K/F)$ .  
 (Note: the field of characteristic 0 or finite field are perfect.)

1)  $F = \mathbb{Q}$ .

By the Eisenstein criterion, all coefficients of  $x^6+3$  except the leading coefficients are contained in  $3\mathbb{Z}$ , but the square of constant term  $3 \notin 3^2\mathbb{Z}$ , the given polynomial  $x^6+3$  is irreducible over  $\mathbb{Q}$ . The  $x^6+3$  has 6 roots,

$$\sqrt[6]{3} \cdot \omega^{2k+1} \quad (k=0,1,\dots,5) \text{ where } \omega = \exp\left(\frac{i\pi}{6}\right) = \frac{\sqrt{3}}{2} + i\frac{1}{2}.$$

Therefore, the splitting field is  $K = \mathbb{Q}(\sqrt[6]{3} \cdot \omega, \sqrt[6]{3} \cdot \omega^3, \dots, \sqrt[6]{3} \cdot \omega^5)$ ,

clearly  $\mathbb{Q}(\sqrt[6]{3} \cdot \omega) \subseteq K$ , by the definition of generated field.

Meanwhile,  $(\sqrt[6]{3} \cdot \omega)^3 = \sqrt{3} \cdot \omega^3 = \sqrt{3} \cdot \exp(i\pi/2) = \sqrt{3} \cdot i$  and  $\omega^2 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$  implies  $\omega^2 \in \mathbb{Q}(\sqrt[6]{3} \cdot \omega)$ , so  $K = \mathbb{Q}(\sqrt[6]{3} \cdot \omega)$ . Finally, since  $x^6+3$  is irreducible,  $[K:\mathbb{Q}] = 6$ .

Moreover,  $\text{Gal}(K/\mathbb{Q}) \cong S_3$ , because: define two automorphisms on  $K$  as:

let  $\theta = \sqrt[6]{3} \cdot \omega$  and  $\lambda = \omega^2$ . Then,  $\theta^3 = i\sqrt{3}$  and  $\lambda = \frac{1}{2} + \frac{\theta^3}{2}$ . Now, let

$$\sigma: \theta \mapsto \theta \lambda \quad \text{and} \quad \tau: \theta \mapsto \theta \lambda^2.$$

Then,  $\sigma(\lambda) = \sigma\left(\frac{1}{2} + \frac{1}{2}\theta^3\right) = \frac{1}{2} + \frac{1}{2}\theta^3 \lambda^3 = \frac{1}{2} + \frac{1}{2}i\sqrt{3} \cdot \omega^6 = \frac{1}{2}(1 - i\sqrt{3}) = \omega^{10} = \lambda^{-1}$ ,  
 so  $\sigma^2(\theta) = \sigma(\theta \lambda) = \theta \lambda \lambda^{-1} = \theta$ , it means  $\sigma^2 = \text{id}$ .

And  $\tau(\lambda) = \frac{1}{2} + \frac{1}{2}\theta^3 \lambda^6 = \frac{1}{2} + \frac{1}{2}i\sqrt{3} = \lambda$ , so

$$\tau^2(\theta) = \tau(\theta \lambda^2) = \theta \lambda^4, \quad \tau^3(\theta) = \theta \lambda^6 = \theta, \text{ so } \tau^3 = \text{id}.$$

And,  $\sigma\tau(\theta) = \sigma(\theta \lambda^2) = \theta \cdot \lambda^{-1} = \theta \cdot \lambda^2$ ,  $(\sigma\tau)^2(\theta) = \sigma\tau(\theta \lambda^2) = \theta \lambda^{-3} = \theta$ ,  
 therefore  $\sigma\tau\sigma\tau = \text{id}$ , or  $\sigma\tau\sigma = \tau^{-1}$ . Now,  $\text{Gal}(K/F) = \langle \sigma, \tau \rangle \cong D_6 \cong S_3$ .

Alternatively, if  $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/6\mathbb{Z}$ , the abelian group, then every subgroup is normal. Therefore every intermediate field  $E$  satisfies  $E/F$  is Galois. (In other say, normal)  
 Since  $\sqrt[6]{3} \cdot \omega^{11}, \sqrt[6]{3} \cdot \omega = \sqrt[3]{3} \in K$ , the field  $\mathbb{Q}(\sqrt[3]{3})$  is the intermediate field, but as we know, the minimal polynomial of  $\sqrt[3]{3}$  is  $x^3-3$  over  $\mathbb{Q}$ , but  $\mathbb{Q}(\sqrt[3]{3})$  does not have other roots of  $x^3-3$ . So, it is contradiction.

By the knowledge in the group theory, there are only two group of order 6,  $S_3$  and  $\mathbb{Z}/6\mathbb{Z}$ , so  $\text{Gal}(K/\mathbb{Q}) \cong S_3$ . □

Galois group of  $x^6+3$  over  $F = \mathbb{Q}$  or  $\mathbb{F}_5$  or  $\mathbb{F}_7$ . (UCLA Qual, 2020 Fall)  
 For each case, determine the splitting field  $K$  and evaluate  $[K:F]$ .  
 And, find the Galois group  $\text{Gal}(K/F)$ .  
 (Note: the field of characteristic 0 or finite field are perfect.)

2).  $F = \mathbb{F}_5$

Suppose that  $\alpha \in \overline{\mathbb{F}_5}$  satisfies  $\alpha^2=3$ . Then,  $(\alpha^2)^3+3=3\alpha=0 \pmod 5$ ,  
 therefore  $x^2-3$  divides  $x^6+3$ , and  $x^2-3$  is irreducible over  $\mathbb{F}_5$ ,  
 because substituting  $x=0,1,\dots,4$ ,  $x^2-3 \neq 0$ , by casework. ( $\{x^2 | x \in \mathbb{F}_5\} = \{0,1,4\}$ )  
 Let  $K = \mathbb{F}_5(\alpha)$ . Since  $x^2-3$  is the minimal polynomial of  $\alpha$ ,  $[K:\mathbb{F}_5] = 2$ .  
 Given  $a+b\alpha \in K$  ( $a,b \in \mathbb{F}_5$ ), observe that

$$(a+b\alpha)^6 = \underbrace{(a+b\alpha)^5}_{\text{Frobenius automorphism which fixes } \mathbb{F}_5} \cdot (a+b\alpha) = (a+b\alpha^5) \cdot (a+b\alpha) = (a+b\alpha)(a-b\alpha) = a^2 + \underbrace{2b^2}_{=-3} = 2$$

'if and only 'if':

$$\begin{cases} a=0 \\ b=\pm 1 \end{cases} \quad \text{or} \quad \begin{cases} a=\pm 2 \\ b=\pm 2 \end{cases} \quad \Leftrightarrow \quad \begin{cases} a=2 \text{ or } 3 \\ b=2 \text{ or } 3 \end{cases}$$

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So, the six distinct roots are contained in  $K$ , so  $K$  is splitting field of  $x^6+3$ ,  
 and clearly it is separable, therefore  $K/\mathbb{F}_5$  is Galois, with the Galois group

$$\text{Gal}(K/\mathbb{F}_5) = \langle \alpha \mapsto \alpha^5 \rangle \cong \mathbb{Z}_2.$$

In actually,  $K = \mathbb{F}_{5^2}$ .

Galois group of  $x^6+3$  over  $F = \mathbb{Q}$  or  $\mathbb{F}_5$  or  $\mathbb{F}_7$ . (UCLA QQual, 2020 Fall)  
For each case, determine the splitting field  $K$  and evaluate  $[K:F]$ .  
And, find the Galois group  $\text{Gal}(K/F)$ .  
(Note: the field of characteristic 0 or finite field are perfect.)

3).  $F = \mathbb{F}_7$

Galois group of  $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))$  over  $\mathbb{Q}$

First, since  $x = \sqrt{2+\sqrt{2}} \Rightarrow x^4 - 4x^2 + 2 = 0$ , the degree of minimal polynomial of  $\sqrt{2+\sqrt{2}}$  divides 4. And, since  $1 \notin 2\mathbb{Z}$ ,  $0, 4, 2 \in 2\mathbb{Z}$  but  $2 \in 2^2\mathbb{Z}$ , so the Eisenstein criterion gives that the given polynomial is irreducible, therefore  $x^4 - 4x^2 + 2$  is exactly the minimal polynomial of  $\sqrt{2+\sqrt{2}}$ .

By the fundamental theorem of Algebra, we can factor  $x^4 - 4x^2 + 2$  as linear factors, as

$$\begin{aligned} x^4 - 4x^2 + 2 &= (x^2 - (+2 + \sqrt{2}))(x^2 - (+2 - \sqrt{2})) \\ &= (x - \sqrt{2+\sqrt{2}})(x + \sqrt{2+\sqrt{2}})(x - \sqrt{2-\sqrt{2}})(x + \sqrt{2-\sqrt{2}}) \end{aligned}$$

To show that  $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))$  is splitting field of  $x^4 - 4x^2 + 2$ ,

enough to show that  $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\sqrt{2+\sqrt{2}})$ .

Observe that:  $\sqrt{2} = (\sqrt{2+\sqrt{2}})^2 - 2$  and

$$(\sqrt{2+\sqrt{2}}) \cdot (\sqrt{2-\sqrt{2}}) = \sqrt{2}, \text{ so } \sqrt{2-\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2+\sqrt{2}}} = \sqrt{2+\sqrt{2}} - \frac{2}{\sqrt{2+\sqrt{2}}}$$

therefore  $\sqrt{2-\sqrt{2}} \in \mathbb{Q}(\sqrt{2+\sqrt{2}})$ . And, the formal derivative  $4x^3 - 8x = 4x(x^2 - 2)$  has roots as  $0, \pm\sqrt{2}$ , so that  $x^4 - 4x^2 + 2$  is separable.

This proves  $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))/\mathbb{Q}$  is Galois, and has degree of 4.

Now, consider an automorphism on  $(\mathbb{Q}(\sqrt{2+\sqrt{2}}))$  as

$$\sigma: \underset{\alpha''}{\sqrt{2+\sqrt{2}}} \mapsto \underset{\beta}{\sqrt{2-\sqrt{2}}}$$

$$\begin{aligned} \alpha^4 - 4\alpha^2 + 2 &= 0 / \beta = \\ \alpha^3 - 4\alpha &= -\frac{2}{\alpha} \\ -\frac{2}{\alpha} + \beta &= 0 \end{aligned}$$

Then,  $\beta = \alpha^3 - 3\alpha$ , so

$$\sigma^2(\alpha) = \sigma(\beta) = \sigma(\alpha^3 - 3\alpha) = \sigma(\alpha)^3 - 3\sigma(\alpha) = \beta^3 - 3\beta = \beta - \frac{2}{\beta} = -\alpha$$

So  $\sigma$  has order 4, therefore  $\text{Gal}((\mathbb{Q}(\sqrt{2+\sqrt{2}}))/\mathbb{Q}) = \langle \sigma \rangle \cong \mathbb{Z}/4\mathbb{Z}$ .

$\mathbb{Q}(\sqrt{2})$  fixed by  $\langle \sigma^2 \rangle$

$$\text{Note: } \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = \cos^2 \alpha + \cos^2 \alpha - 1 \Rightarrow \cos^2 \alpha = \frac{1 + \cos 2\alpha}{2} = 1 - \sin^2 \alpha$$

$$\text{So, } \cos \frac{\pi}{8} = \sqrt{\frac{1}{2} + \frac{1}{2} \frac{\sqrt{2}}{2}} = \frac{1}{2} \sqrt{2 + \sqrt{2}} \text{ and } \sin \frac{\pi}{8} = \frac{1}{2} \sqrt{2 - \sqrt{2}}$$

Biquadratic Criterion:  $2^2 - 2 = 2$  is not square.



Galois group of  $\mathbb{Q}(\sqrt{p} + \sqrt{q})$  over  $\mathbb{Q}$ , where  $p$  and  $q$  are distinct prime (UCLA Spring 2023)

Galois group of  $\mathbb{Q}(x)/\mathbb{Q}(x^2)$

$\Gamma_{\text{Galois group of } \mathbb{Q}(x,y)/\mathbb{Q}(xy)}$